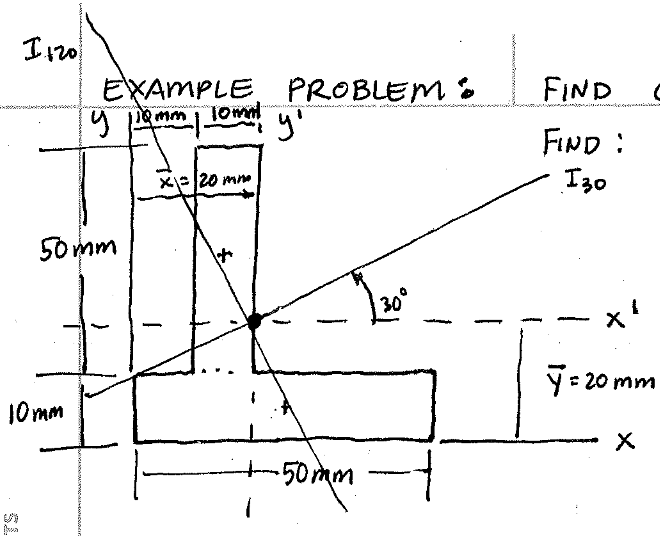
EXAMPLE PROBLEM: FIND CENTROID $\bar{x}, \bar{y}$; $I_{x'}, I_{y'}, I_{x'y'}$;

FIND: $I_{30}, I_{120}, I_{xy30}$; $I_{MAX}, I_{MIN}, \theta_P$



CENTROID: $A = 2(10)(50) = 1000\ m^2$

$$\bar{y} = \frac{5(10)(50) + 35(10)(50)}{1000} = \boxed{20\ mm = \bar{y}}$$

$$\bar{x} = \frac{25(10)(50) + 15(10)(50)}{1000} = \boxed{20\ mm = \bar{x}}$$

ABOUT CENTROIDAL AXES:

$$I_{x'} = \Sigma(\bar{I} + Ad^2) = \left[\tfrac{1}{12}(10)(50)^3 + 10(50)(15)^2\right] \left[\tfrac{1}{12}(50)(10)^3 + 10(50)(15)^2 +\right] = \boxed{333.3 \cdot 10^3\ mm^4}$$

$$I_{y'} = \left[\tfrac{1}{12}(10)(50^3) + (10)(50)(5)^2\right] + \left[\tfrac{1}{12}(50)(10)^3 + 10(50)5^2\right] = \boxed{133.3 \cdot 10^3\ mm^4 = I_{y'}}$$

$$I_{x'y'} = \Sigma(\bar{I} + A d_x d_y) = \left[0 + 10(50)(+5)(-15)\right] + \left[0 + 10(50)(-5)(+15)\right] = \boxed{-75 \cdot 10^3\ mm^4}$$

$$\boxed{I_\theta = I_x \cos^2\theta + I_y \sin^2\theta - 2I_{xy} \sin\theta\cos\theta} \qquad \boxed{I_{xy_\theta} = I_{xy}\cos 2\theta + \left(\frac{I_x - I_y}{2}\right)\sin 2\theta}$$

$$I_{30} = (333.3 \cdot 10^3)(\underbrace{\cos^2 30}_{.75}) + (133.3 \cdot 10^3)(\underbrace{\sin^2 30}_{.25}) - 2(-75 \cdot 10^3)\sin 30\cos 30$$

$$\boxed{I_{30} = 348.3 \cdot 10^3\ mm^4}$$

$$I_{120} = (333.3 \cdot 10^3)\underbrace{\cos^2 120}_{.25} + 133.3 \cdot 10^3 \underbrace{\sin^2 120}_{.75} - 2(-75 \cdot 10^3)\sin 120\cos 120$$

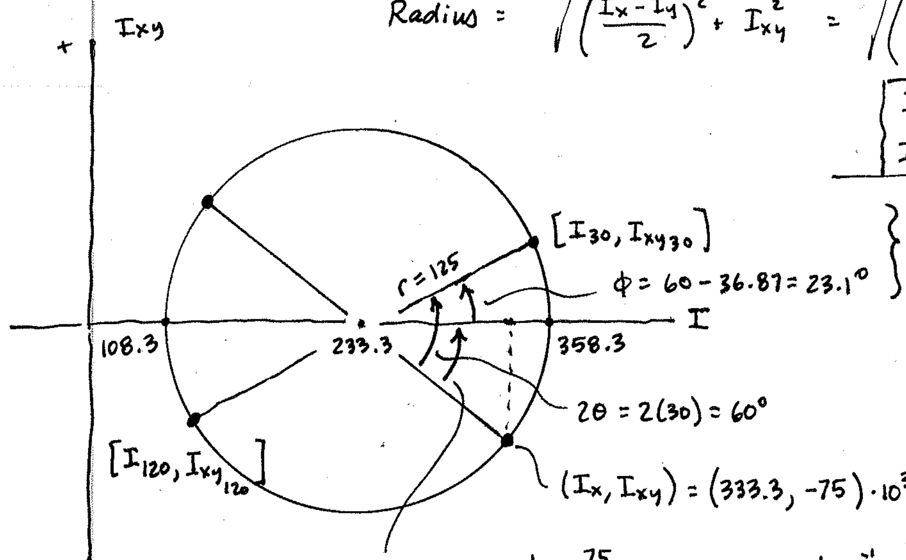
$$\boxed{I_{120} = 118.3 \cdot 10^3\ mm^4}$$

$$I_{xy_{30}} = (-75 \cdot 10^3)\cos 2(30) + \left(\frac{333.3 \cdot 10^3 - 133.3 \cdot 10^3}{2}\right)\sin 2(30) \Rightarrow \boxed{I_{xy_{30}} = 49.1 \cdot 10^3\ mm^4}$$

MOHR'S CIRCLE: CENTER $= I_{AVG} = \dfrac{333.3 + 133.3}{2} = 233.3 \cdot 10^3\ mm^4$

$$\text{Radius} = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2} = \sqrt{\left(\frac{333.3 - 133.3}{2} \cdot 10^3\right)^2 + (-75 \cdot 10^3)^2} = 125 \cdot 10^3\ mm^4$$

$$\boxed{\begin{aligned} I_{MAX} &= 233.3 + 125 = 358.3 \cdot 10^3\ mm^4 \\ I_{MIN} &= 233.3 - 125 = 108.3 \cdot 10^3\ mm^4 \end{aligned}}$$



$I_{30} = 233.3 + 125\cos 23.1$

$$\boxed{I_{30} = 348.25 \cdot 10^3\ mm^4}$$

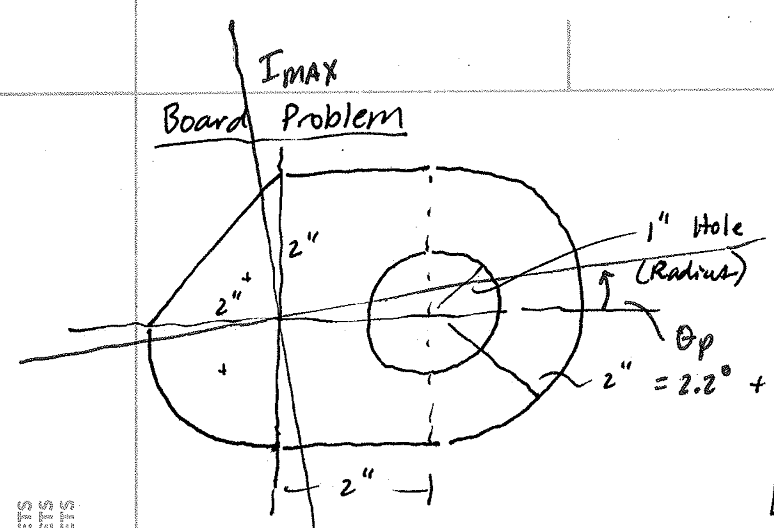
$$I_{xy_{30}} = 125\sin 23.1 = \boxed{+49.1 \cdot 10^3\ mm^3 = I_{xy_{30}}}$$

$$2\theta_P = \tan^{-1}\frac{75}{333.3 - 233.3} = \tan^{-1}\frac{I_{xy}}{I_x - I_{AVG}}$$

$$2\theta_P = 36.87°$$

$$\boxed{\theta_P = 18.4°}$$

Board Problem



Find: $I_x, I_y, I_{xy}, J, k_x, k_y, I_{MAX}, I_{MIN}, \theta_p$

$$I_x = \Sigma(\bar{I} + Ad^2) = \frac{1}{12}(2)(2)^3_{\Delta} + \frac{\pi}{16}2^4_{\square} + \frac{1}{12}(2)(4)^3 + \frac{\pi}{8}(2)^4 - \frac{\pi}{4}(1)^4$$

$$\boxed{I_x = 20.64 \text{ in}^4}$$

$$I_y = \Sigma(\bar{I} + Ad^2) = \frac{1}{12}(2)(2)^3 + \frac{\pi}{16}(2)^4 + \frac{1}{3}(4)(2)^3 + \left[.1098(2)^4 + \frac{\pi}{2}(2)^2\left(2 + \frac{4(2)}{3\pi}\right)^2\right]$$

$$- \left[\frac{\pi}{4}(1)^4 + \pi(1)^2(2)^2\right] = \boxed{54.54 \text{ in}^4 = I_y}$$

$$I_{xy} = \Sigma(\bar{I}_{xy} + A d_x d_y) = \left[+\frac{1}{72}(2)^2(2)^2 + \frac{1}{2}(2)(2)\left(-\frac{2}{3}\right)\left(+\frac{2}{3}\right)\right]$$

$$+ \left[-.01647(2)^4 + \frac{\pi}{4}(2)^2\left(\frac{-4(2)}{3\pi}\right)\left(\frac{-4(2)}{3\pi}\right)\right] + 0 + 0 + 0$$

$$= -.6667 + 2.00 \Rightarrow \boxed{I_{xy} = 1.333 \text{ in}^4}$$

I_{xy} Note: + and − values of d_x + d_y and + and − values of Δ and $\square$

$$J_o = I_x + I_y = 20.64 + 54.54 = \boxed{75.18 \text{ in}^4 = J_o}$$

$$\text{Area} = \frac{1}{2}(2)(2) + \frac{\pi}{4}(2)^2 + 2(4) + \frac{\pi}{2}(2)^2 - \pi(1)^2 = \boxed{16.28 \text{ in}^2 = A}$$

$$k_x = \sqrt{I_x/A} = \sqrt{\frac{20.64}{16.28}} = \boxed{1.126 \text{ in} = k_x}$$

$$k_y = \sqrt{\frac{I_y}{A}} = \sqrt{\frac{54.54}{16.28}}$$

$$\boxed{k_y = 1.830 \text{ in}}$$

$$I_{\substack{MAX \\ MIN}} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$

$$= \left(\frac{20.64 + 54.54}{2}\right) \pm \sqrt{\left(\frac{20.64 - 54.54}{2}\right)^2 + 1.333^2}$$

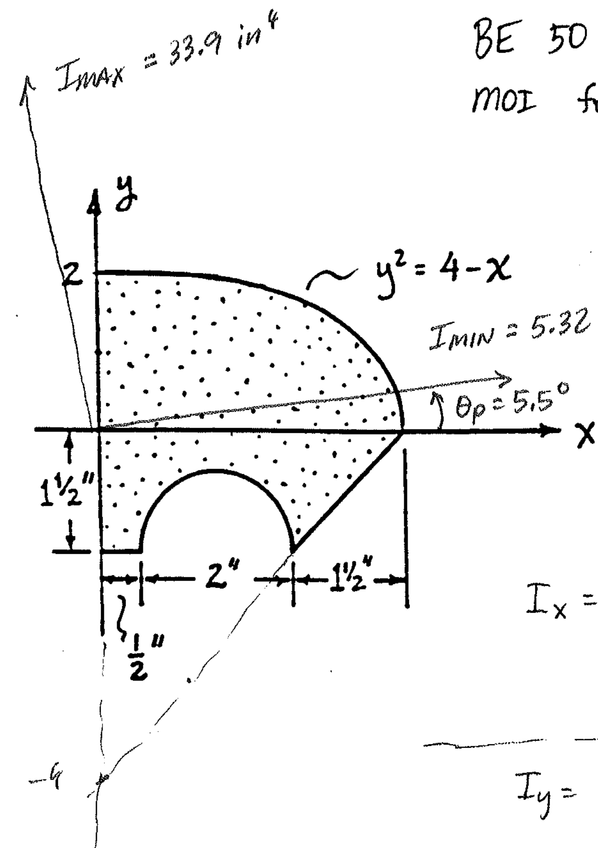
$$= 37.59 \pm 17.0$$

$$\boxed{\begin{aligned} I_{MAX} &= 54.59 \text{ in}^4 \\ I_{MIN} &= 20.59 \text{ in}^4 \end{aligned}}$$

$$2\theta_p = \tan^{-1}\frac{-I_{xy}}{\frac{I_x - I_y}{2}} = \tan^{-1}\frac{-1.33}{\frac{20.64 - 54.54}{2}} = 4.49°$$

$$\boxed{\theta_p = 2.24°}$$

BE 50 Homework
MOI for a Composite Area



Please find: $I_x, I_y, J_o, I_{xy}, \theta_p, I_{\substack{MAX\\MIN}}$

Answers: $I_x = 5.574 \text{ in}^4$

$I_y = 33.66 \text{ in}^4$

$I_{xy} = 2.735 \text{ in}^4$

[Just for integrated area: $I_x = 4.267 \text{ in}^4$, $I_y = 19.51 \text{ in}^4$, $I_{xy} = 5.33 \text{ in}^4$]

$$I_x = \int_0^2\int_0^{4-y^2} y^2\,dx\,dy = \int_0^2 y^2(4-y^2)\,dy = \int_0^2 (4y^2 - y^4)\,dy = \left(\tfrac{4}{3}y^3 - \tfrac{1}{5}y^5\right)\Big|_0^2$$

$$= 32\left(\tfrac{1}{3} - \tfrac{1}{5}\right) = \frac{64}{15} = \boxed{4.267 \text{ in}^4 = I_x}$$

$$I_y = \int_0^2\int_0^{4-y^2} x^2\,dx\,dy = \int_0^2 \tfrac{1}{3}(4-y^2)^3\,dy$$

$(4-y^2)^2 = (16 - 8y^2 + y^4)$

$(16 - 8y^2 + y^4)(4 - y^2)$

$= (64 - 48y^2 + 12y^4 - y^6)$

$$= \int_0^2 \tfrac{1}{3}(64 - 48y^2 + 12y^4 - y^6)\,dy$$

$$= \tfrac{1}{3}\left(64y - 16y^3 + \tfrac{12}{5}y^5 - \tfrac{1}{7}y^7\right)\Big|_0^2 = \tfrac{1}{3}\left(\cancel{128} - 16(\cancel{8}) + \tfrac{12}{5}(32) - \tfrac{1}{7}(128)\right)$$

$$= \frac{32}{3}\left(\frac{84}{35} - \frac{20}{35}\right) = \frac{32}{3}\left(\frac{64}{35}\right) = \boxed{19.505 \text{ in}^4 = I_y}$$

$y = -4 + x$

$x = y + 4$

$$I_{xy} = \int_0^2\int_0^{4-y^2} xy\,dx\,dy = \int_0^2 \tfrac{1}{2}y(4-y^2)^2\,dy = \int_0^2 \tfrac{1}{2}(16y - 8y^3 + y^5)\,dy$$

$$= \tfrac{1}{2}\left(8y^2 - 2y^4 + \tfrac{1}{6}y^6\right)\Big|_0^2 = \tfrac{1}{2}\left(32 - 32 + \tfrac{1}{6}(64)\right) = \frac{64}{12}$$

$$\boxed{I_{xy} = 5.33 \text{ in}^4}$$

$$I_x = 4.267 +_{\square}\left[\tfrac{1}{3}(2.5)(1.5)^3\right] +_{\triangledown}\left[\tfrac{1}{12}(1.5)(1.5)^3\right] -\left[.1098(1)^4 + \tfrac{\pi(1)^2}{2}\left(1.5 - \tfrac{4(1)}{3\pi}\right)^2\right]$$

$$= 4.267 + 2.813 + 0.4219 - 1.927 = \boxed{5.574 \text{ in}^4 = I_x}$$

$$I_y = 19.505 +_{\square}\left[\tfrac{1}{3}(1.5)(2.5)^3\right] +_{\triangledown}\left[\tfrac{1}{36}(1.5)(1.5)^3 + \tfrac{1}{2}(1.5)(1.5)(2.5 + .5)^2\right] - \left[\tfrac{\pi}{8}(1)^4 + \tfrac{\pi(1)^2}{2}(1.5)^2\right]$$

$$= 19.505 + 7.813 + 10.266 - 3.927 = \boxed{33.66 \text{ in}^4 = I_y}$$

$$I_{xy} = 5.33 +_{\square}\left[0 + (1.5)(2.5)(1.25)(-.75)\right] +_{\triangledown}\left[+\tfrac{1}{72}(1.5)^4 + \tfrac{1}{2}(1.5)^2(3.0)(-.5)\right] - \left[0 + \tfrac{\pi}{2}(1)^2(1.5)\left(-1.5 + \tfrac{4}{3\pi}\right)\right]$$

$$= 5.33 + (-3.516) + (-1.617) - (-2.534) = \boxed{2.732 \text{ in}^4 = I_{xy}}$$

$$\tan 2\theta_p = \frac{-I_{xy}}{(I_x - I_y)/2} = \frac{-2.732}{(5.574 - 33.66)/2} \Rightarrow 2\theta_p = 11.0° \quad \boxed{\theta_p = 5.5°}$$

$$I_{\substack{MAX\\MIN}} = I_{AVG} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$

$$= 19.62 \pm 14.30$$

$$\boxed{I_{\substack{MAX\\MIN}} = 33.92 \text{ in}^4,\ 5.32 \text{ in}^4}$$

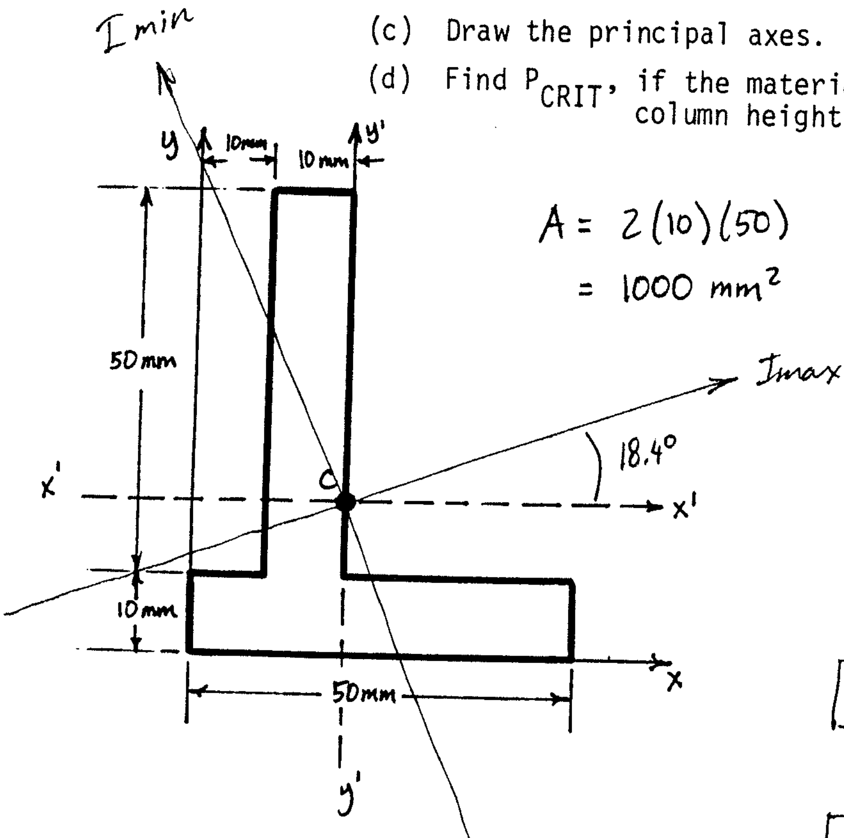
4. (25 points) For the area shown below, $I_x = 733.3 \times 10^3\ mm^4$, $I_y = 533.3 \times 10^3\ mm^4$, $I_{xy} = 325 \times 10^3\ mm^4$, $\bar{x} = 20$ mm, and $\bar{y} = 20$ mm.

Please find: (a) $\bar{I}_{x'}$, $\bar{I}_{y'}$, and $\bar{I}_{x'y'}$ at the centroid.

(b) θ_p, I_{MAX}, I_{MIN}

(c) Draw the principal axes.

(d) Find P_{CRIT}, if the material is aluminum, and the intended column height is 12 m.



$$A = 2(10)(50)$$
$$= 1000\ mm^2$$

EULER'S EQUATION:

$$P_{CRIT} = \frac{\pi^2 E I}{L^2}$$

$E_{Aluminum} = 70 \times 10^9\ N/m^2$

$= 10 \times 10^6$ psi

$$\bar{I}_{x'} = I_x - Ad^2$$
$$= (733.3 \times 10^3) - (10^3)(20)^2$$
$$\boxed{\bar{I}_{x'} = 333.3 \cdot 10^3\ mm^4}$$

$$\bar{I}_{y'} = 533.3 \cdot 10^3 - (10^3)(20)^2$$
$$\boxed{\bar{I}_{y'} = 133.3 \cdot 10^3\ mm^4}$$

$$\bar{I}_{x'y'} = 325 \cdot 10^3 - (10^3)(20)(20)$$
$$\boxed{\bar{I}_{x'y'} = -75. \times 10^3\ mm^4}$$

$$I_{\substack{MAX\\MIN}} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$
$$= 233.3 \times 10^3 \pm \sqrt{1.563 \cdot 10^{10}}$$
$$= 233.3 \cdot 10^3 \pm 125 \cdot 10^3$$
$$\boxed{I_{MAX} = 358.3 \cdot 10^3\ mm^4}$$
$$\boxed{I_{MIN} = 108.3 \cdot 10^3\ mm^4}$$

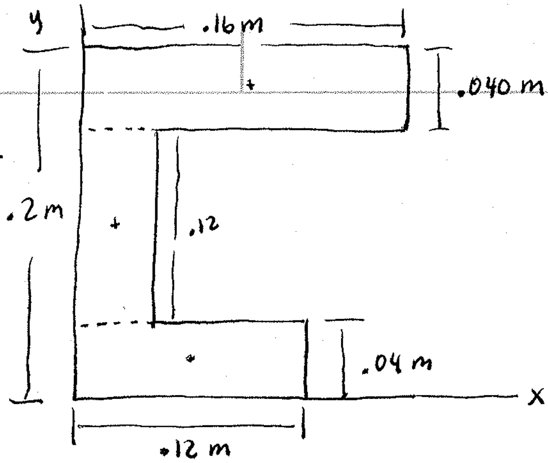
$$\tan 2\theta_p = \frac{-I_{xy}}{\frac{I_x - I_y}{2}}$$
$$= \frac{-(-75 \cdot \cancel{10^3})}{\frac{(333.3 - 133.3)}{2} \cdot \cancel{10^3}} = 0.75$$

$$2\theta_p = 36.9°$$
$$\boxed{\theta_p = 18.4°}$$

$$P_{CRIT} = \frac{\pi^2 (70 \cdot \cancel{10^9}\ N/m^2)(108.3 \cdot \cancel{10^3}\ mm^4)\left(\frac{m}{\cancel{10^3}\ mm}\right)^{\cancel{4}}}{(12)^2}$$

$$\boxed{P_{CRIT} = 519.6\ N}$$

B-35, 36, 37



$$I_y = \Sigma(\bar{I} + Ad^2) = \Sigma\left(\tfrac{1}{3}bh^3\right) = \left[\tfrac{1}{3}(.04)(.12)^3\right] + \left[\tfrac{1}{3}(.04)(.16)^3\right] + \left[\tfrac{1}{3}(.12)(.04)^3\right]$$

$$I_y = 23.04 \cdot 10^{-6} + 54.61 \cdot 10^{-6} + 2.56 \cdot 10^{-6}$$

$$\boxed{\begin{aligned} I_y &= 80.21 \cdot 10^{-6}\ m^4 \\ &= 80.21 \cdot 10^{6}\ mm^4 \end{aligned}}$$

$$I_x = \Sigma(\bar{I} + Ad^2) = \left[\tfrac{1}{3}(.12)(.04)^3\right] + \left[\tfrac{1}{12}(.04)(.12)^3 + (.04)(.12)(.1)^2\right] + \left[\tfrac{1}{12}(.16)(.04)^3 + (.16)(.04)(.18)^2\right]$$

$$I_x = 2.56 \cdot 10^{-6} + 53.76 \cdot 10^{-6} + 208.2 \cdot 10^{-6}$$

$$\boxed{\begin{aligned} I_x &= 264.5 \cdot 10^{-6}\ m^4 \\ &= 264.5 \cdot 10^{6}\ mm^4 \end{aligned}}$$

Area: $A = (.04)(.12) + (.12)(.04) + .16(.04)$

$$= .04(.12 + .12 + .16)$$

$$A = .04(.4) = 16.0 \cdot 10^{-3}\ m^2$$

$$k_x = \sqrt{I_x/A} = \sqrt{\frac{264.5 \cdot 10^{-6}}{16 \cdot 10^{-3}}} = \boxed{0.1286\ m = k_x}$$

$$k_y = \sqrt{I_y/A} = \sqrt{\frac{80.21 \cdot 10^{-6}}{16 \cdot 10^{-3}}} = \boxed{0.0708\ m = k_y}$$

$$I_{xy} = \Sigma(\bar{I}_{xy} + A d_x d_y)$$

$$= \left[0 + (.04)(.12)(.02)(.06)\right] + \left[0 + (.04)(.12)(.02)(.1)\right] + \left[0 + (.04)(.16)(.18)(.08)\right]$$

$$= 5.76 \cdot 10^{-6} + 9.6 \cdot 10^{-6} + 92.16 \cdot 10^{-6}$$

$$\boxed{I_{xy} = 107.52 \cdot 10^{-6}\ m^4}$$